Master Learning Roadmap: AI for Cybersecurity Engineering

Executive Summary

Your Mission: Build AI-augmented security tools to prove value at work, reduce incident load, and create commercializable products.

Timeline: 12 weeks to working prototype → 24 weeks to commercialization

Hardware: 198GB RAM, 32 CPU cores, NVIDIA GPU (optimal for local deployment)

Key Constraint: All work must be done locally (air-gapped, proprietary security data)

The Strategic Learning Arc

Phase 1: SURVIVAL (Weeks 1-6) - Prove Your Worth

Goal: Create user-facing chatbot that reduces tickets by 30%
Impact: Job security, immediate visibility
Investment: ~60 hours over 6 weeks

Phase 2: DEPTH (Weeks 7-12) - Build Engineering Excellence

Goal: Production-grade RAG system with monitoring
Impact: Portfolio-worthy project, technical credibility
Investment: ~80 hours over 6 weeks

Phase 3: COMMERCIALIZATION (Weeks 13-24) - Revenue Generation

Goal: Packaged security tools for external sales
Impact: Income stream, career optionality
Investment: ~100 hours over 12 weeks

How the 9 Subjects Map to Your Goals

Foundation Layer (Weeks 1-3)

Subject 9: Local LLM Deployment

Why First: You can't build anything without models running locally
Your Application: Deploy Llama 3.1 70B on your hardware
Security Angle: Air-gapped deployment, no data leaving network
Deliverable: Ollama running, basic inference working

Subject 1: Prompt Engineering

Why Second: Transform local models into reliable tools
Your Application: Security-specific prompts (incident classification, log analysis)
Security Angle: Prompt injection defense for user-facing chatbot
Deliverable: Prompt library for common security tasks

Subject 4: Structured Output Generation

Why Third: Chatbot must return parseable data (JSON, not prose)
Your Application: Extract IOCs from logs, classify incidents into categories
Security Angle: Structured audit trails for compliance
Deliverable: Pydantic schemas for security outputs

Application Layer (Weeks 4-8)

Subject 5: Production-Grade RAG

Why Now: Your security docs become queryable knowledge base
Your Application: "Ask anything" interface for playbooks, policies, past incidents
Security Angle: Local vector DB (ChromaDB), no cloud embeddings
Deliverable: RAG system over your security documentation

Subject 2: Data Engineering for LLM Apps

Why Now: Optimize document chunking and embedding pipeline
Your Application: Process security logs, playbooks, vulnerability reports
Security Angle: Incremental updates without full reindex
Deliverable: Efficient data pipeline for large log volumes

Subject 3: Token-Level Cost Optimization

Why Now: Local resources are finite (RAM, GPU)
Your Application: Model routing (8B for simple, 70B for complex)
Security Angle: Resource monitoring, prevent OOM crashes
Deliverable: Multi-model architecture with intelligent routing

Production Layer (Weeks 9-16)

Subject 6: LLM Observability & Debugging

Why Now: Production systems need monitoring
Your Application: Trace incident analysis workflows, log all AI decisions
Security Angle: Audit trails for compliance, incident response playbooks
Deliverable: Monitoring dashboard with alerts

Subject 7: LLM Evaluation

Why Now: Prove your tools work with metrics
Your Application: Measure incident classification accuracy, false positive rates
Security Angle: Security-specific metrics (threat detection recall)
Deliverable: Evaluation framework with benchmarks

Subject 8: Agent Frameworks

Why Now: Automate multi-step security workflows
Your Application: Automated threat hunting, incident investigation agents
Security Angle: AI Security Analyst (your commercialization target)
Deliverable: ReAct-based security agent

Your Three Deliverables (Mapped to Subjects)

Deliverable 1: User-Facing Security Chatbot (Week 1-6)

Business Value: Reduces ticket volume, frees up your time
Subjects Used:
- Subject 9 (Local deployment)
- Subject 1 (Prompt engineering)
- Subject 4 (Structured outputs)

Architecture:

```
User Question
    ↓
Intent Classifier (Subject 1 prompts)
    ↓
Route to Handler:
  - Password Reset → Auto-resolve (Subject 4 structured output)
  - Account Lock → Diagnostic workflow (Subject 1 prompts)
  - Complex Query → RAG lookup (Subject 5)
    ↓
Response (JSON or natural language)
    ↓
Audit Log (Subject 6 observability)
```

Key Features:
- Handles 5 most common ticket types automatically
- Structured JSON responses for downstream automation
- Prompt injection defense (Subject 1 security patterns)
- Local deployment (Subject 9 air-gapped setup)

Success Metrics (Subject 7):
- 30% reduction in tickets to you
- <5 second response time (Subject 3 optimization)
- 90%+ accuracy on known questions (Subject 7 evaluation)

Deliverable 2: Security Knowledge Base (RAG System) (Week 4-8)

Business Value: Institutional knowledge becomes accessible to entire team
Subjects Used:
- Subject 5 (RAG architecture)
- Subject 2 (Data engineering)
- Subject 9 (Local embeddings)

Architecture:

```
Security Documents
    ↓
Chunking Pipeline (Subject 2)
  - Semantic chunking for playbooks
  - Fixed-size for logs
  - Overlap optimization
```

```
    ↓
Embedding Generation (Subject 9 local)
  - sentence-transformers (bge-large-en-v1.5)
  - Batch processing on GPU
  - Caching for efficiency
    ↓
Vector Database (Subject 5)
  - ChromaDB local mode
  - Metadata filtering (Subject 5 reranking)
  - Hybrid search (vector + keyword)
    ↓
Query Processing
  - Query expansion (Subject 5)
  - Top-k retrieval → Reranking (Subject 5)
  - Context optimization (Subject 3)
    ↓
Answer Generation
  - Local LLM (Subject 9)
  - Structured citations (Subject 4)
  - Audit logging (Subject 6)
```

Document Sources:
1. **Security Playbooks**: Incident response procedures, threat hunting guides
2. **Internal Policies**: Access control policies, compliance requirements
3. **Past Incidents**: Lessons learned, postmortems, root cause analyses
4. **Vulnerability Reports**: CVE databases, patch management docs
5. **Training Materials**: Security awareness content, onboarding guides

Key Features:
- Semantic search: "How do we handle golden ticket attacks?" → Retrieves Kerberos playbook
- Cited responses: Shows which documents were used (Subject 4 structured sources)
- Incremental updates: Add new docs without full reindex (Subject 2)
- Access control: Filter by clearance level (Subject 5 metadata filtering)

Success Metrics (Subject 7):
- Answer relevance: 95%+ (LLM-as-judge evaluation)
- Citation accuracy: 90%+ (answers grounded in retrieved docs)
- Retrieval recall: Top-5 contains relevant doc 85%+ of time
- Query latency: <3 seconds end-to-end

Deliverable 3: AI Security Analyst Agent (Week 9-16)

Business Value: Automate threat hunting, commercializable product
Subjects Used:
- Subject 8 (Agent frameworks)
- Subject 1, 4, 5 (Tools for agent)
- Subject 6, 7 (Production ops)

Architecture:

```
Security Alert
    ↓
Agent Reasoning Loop (Subject 8 ReAct)
    ↓
Tool Selection:
  1. Log Analysis Tool (Subject 4 structured extraction)
     - Extract: Source IP, timestamps, user, action
     - Classify: Benign vs suspicious

  2. Threat Intelligence Tool (Subject 5 RAG)
     - Query: "Known attacks matching this pattern?"
     - Retrieve: Historical incidents, CVE data

  3. Network Context Tool (Subject 4 structured API)
     - Query: User behavior baseline, network topology
     - Return: Is this IP/user expected in this context?

  4. Escalation Tool (Subject 4 structured output)
     - Generate: Incident ticket with priority, summary, evidence
     - Route: Human analyst or auto-remediate
    ↓
Multi-Step Investigation
  - Observation: "User failed 20 logins from new IP"
  - Reasoning: "Check if user traveling, VPN expected, or compromised"
  - Action: Query user travel calendar, VPN logs
  - Observation: "No travel scheduled, IP is suspicious"
  - Reasoning: "High probability of credential compromise"
  - Action: Escalate to human with evidence
    ↓
Output: Investigation Report + Recommended Actions
    ↓
Audit Trail (Subject 6)
```

Tools the Agent Uses:
1. **SIEM Query Tool**: Search logs (Splunk, ELK, etc.)
2. **Threat Intel Tool**: RAG over CVE databases, threat reports
3. **Asset Inventory Tool**: Query CMDB for context
4. **User Behavior Tool**: Baseline normal vs anomalous activity
5. **Remediation Tool**: Disable account, block IP, reset password

Key Features:
- Autonomous investigation up to confidence threshold (Subject 8 early stopping)
- Multi-tool orchestration: Uses 3-5 tools per investigation (Subject 8)
- Human-in-the-loop: Escalates when confidence < 80% (Subject 8 control)
- Cost-aware: Uses 8B model for simple tools, 70B for reasoning (Subject 3)

Success Metrics (Subject 7):
- Task completion rate: 70%+ investigations completed without human
- Tool usage accuracy: 95%+ correct tool selections
- False positive rate: <5% (doesn't escalate noise)
- Investigation efficiency: 5x faster than manual (minutes vs hours)

Your 12-Week Learning Schedule

Weeks 1-2: Foundation (Subjects 9, 1)

Focus: Local deployment + prompt engineering
Deliverable: Basic security chatbot (handles 3 question types)

Learning Activities:
- Subject 9: Model selection, quantization, Ollama setup
- Subject 1: Prompt patterns, failure modes, iteration workflow
- Practice: Security-specific prompts (incident classification, log analysis)

Time Investment: 20 hours (10 hours/week)

Output: Working chatbot that answers:
1. "How do I reset my password?"
2. "Why is my account locked?"
3. "What is a phishing email?"

Weeks 3-4: Structured Outputs (Subjects 4, 1)

Focus: Reliable data extraction
Deliverable: Log parser with JSON output

Learning Activities:
- Subject 4: Pydantic schemas, validation, error handling
- Subject 1: Prompts for structured outputs
- Practice: Extract IOCs from security logs

Time Investment: 20 hours

Output: Tool that parses unstructured logs into:

```
{
  "timestamp": "2024-02-10T14:32:01Z",
  "source_ip": "192.168.1.100",
  "event_type": "failed_login",
  "user": "jsmith",
  "risk_score": 0.85,
  "iocs": ["192.168.1.100"],
  "recommended_action": "monitor"
}
```

Weeks 5-6: RAG System (Subjects 5, 9)

Focus: Security knowledge base
Deliverable: Queryable documentation system

Learning Activities:
- Subject 5: RAG architecture, retrieval strategies, reranking
- Subject 9: Local embeddings, ChromaDB setup
- Practice: Index your security playbooks

Time Investment: 20 hours

Output: System that answers:
- "What's the procedure for Kerberos ticket compromise?"
- "Show me past incidents involving credential stuffing"
- "What are our password policy requirements?"

Weeks 7-8: Data Pipeline (Subjects 2, 3)

Focus: Scale and optimize
Deliverable: Efficient document processing

Learning Activities:
- Subject 2: Chunking strategies, embedding pipelines, versioning
- Subject 3: Resource optimization, model routing, caching
- Practice: Process 10,000+ log entries efficiently

Time Investment: 20 hours

Output: Pipeline that:
- Chunks documents semantically (not fixed-size)
- Generates embeddings in batches (GPU-accelerated)
- Incremental updates (only new docs)
- Routes queries to 8B or 70B based on complexity

Weeks 9-10: Observability (Subjects 6, 7)

Focus: Production monitoring
Deliverable: Metrics and alerting

Learning Activities:
- Subject 6: Tracing, logging, error classification
- Subject 7: Evaluation metrics, test datasets
- Practice: Build monitoring dashboard

Time Investment: 20 hours

Output: Dashboard showing:
- Request latency (p50, p95, p99)
- Accuracy over time (daily)
- VRAM usage trends
- Error rates by type
- User satisfaction (thumbs up/down)

Weeks 11-12: Agent System (Subject 8)

Focus: Multi-step automation
Deliverable: Security investigation agent

Learning Activities:
- Subject 8: ReAct pattern, tool design, orchestration
- Integration: Connect tools from Weeks 1-10
- Practice: Automate incident investigation workflow

Time Investment: 20 hours

Output: Agent that:
1. Receives alert: "20 failed logins from IP X"
2. Queries logs: Extract full context (user, time, source)
3. Checks threat intel: Is IP X known malicious?
4. Queries user behavior: Is this normal for this user?
5. Reasoning: Synthesize evidence
6. Action: Escalate to human or auto-remediate

Weeks 13-24: Commercialization (All Subjects)

Focus: Package tools for external sales
Deliverable: Commercial products

Products to Build:
1. **Security Chatbot Starter Kit** ($1-3K/license)
- Pre-trained on common security questions
- Customizable for company policies
- Air-gapped deployment ready

1. **AI Log Analysis Platform** ($5-10K/deployment)
   - Automated IOC extraction
   - Threat classification
   - Integration with SIEM tools

2. **AI Security Analyst Agent** ($20-50K/year SaaS)
   - Automated threat hunting
   - Multi-step investigation workflows
   - Human-in-the-loop approval

Go-to-Market Strategy:
- Target: Mid-size companies (50-500 employees) with small security teams
- Channel: Direct sales, cybersecurity conferences, LinkedIn outreach
- Positioning: "Automate 40% of tier-1 security analyst work"

Subject Dependencies (What to Learn When)

```
Subject 9 (Local Deployment)
    ↓
    Enables ↓
    ↓
Subject 1 (Prompt Engineering) ←→ Subject 4 (Structured Outputs)
    ↓                                     ↓
    Enables ↓                            Enables ↓
    ↓                                     ↓
Subject 5 (RAG) ←──────────────────────┘
    ↓
    Requires ↓
    ↓
Subject 2 (Data Engineering) ←→ Subject 3 (Cost Optimization)
    ↓                                     ↓
    Enables ↓                            Enables ↓
    ↓                                     ↓
Subject 6 (Observability) ←→ Subject 7 (Evaluation)
    ↓                                     ↓
    All Above Enable ↓                    ↓
    ↓←────────────────────────────────────┘
    ↓
Subject 8 (Agent Frameworks)
```

Critical Path (Can't skip or reorder):
1. Subject 9 → Everything else depends on local models running
2. Subject 1 → Foundation for all prompting tasks
3. Subject 5 → Chatbot needs RAG for knowledge base
4. Subject 8 → Agent needs tools from all previous subjects

Flexible (Can reorder based on priorities):
- Subjects 2, 3, 6, 7 can be learned in any order after Subject 5
- Subject 4 can be learned anytime after Subject 1

Weekly Time Investment

Weeks 1-12 (Foundation + Depth)

Learning: 10 hours/week (reading subjects, watching tutorials)
Implementation: 10 hours/week (coding, testing, iterating)
Total: 20 hours/week = 240 hours over 12 weeks

Breakdown by Activity:
- Reading subject docs: 30 hours
- Hands-on coding: 120 hours
- Testing and debugging: 60 hours
- Documentation: 30 hours

Weeks 13-24 (Commercialization)

Product development: 15 hours/week
Marketing/sales: 5 hours/week
Total: 20 hours/week = 240 hours over 12 weeks

Breakdown by Activity:
- Feature development: 120 hours
- Testing and QA: 60 hours
- Documentation and sales materials: 40 hours
- Customer outreach: 20 hours

Success Milestones (How You'll Know You're on Track)

Week 2 Milestone: "Hello World" Working

Checkpoint: Can you query local LLM and get sensible security responses?
Test: Ask model "What is a phishing email?" → Should give accurate answer
If Stuck: Revisit Subject 9 (model setup) and Subject 1 (prompt basics)

Week 6 Milestone: Chatbot Demo-Ready

Checkpoint: Can you show management a working chatbot?
Test: Does it correctly handle 3 question types with 90%+ accuracy?
If Stuck: Revisit Subject 1 (prompt iteration) and Subject 4 (structured outputs)

Week 12 Milestone: RAG System Production-Ready

Checkpoint: Does your team use the knowledge base daily?
Test: Can non-technical users find answers without your help?
If Stuck: Revisit Subject 5 (retrieval quality) and Subject 6 (monitoring)

Week 24 Milestone: First External Customer

Checkpoint: Has someone paid for your tool?
Test: Did they get value without extensive support from you?
If Stuck: Focus on packaging (documentation, easy setup, clear value prop)

Risk Mitigation

Risk 1: "I don't have time" (Most Common Failure)

Symptoms:
- Can't find 20 hours/week
- Work emergencies derail learning
- Family obligations take priority

Mitigation:
- Start with 10 hours/week (slow but steady)
- Front-load weekends (10 hours Saturday/Sunday)
- Use work projects as learning opportunities (expense AI tools to company)

Backup Plan: Extend 12-week plan to 24 weeks (10 hours/week)

Risk 2: "Local models don't work well enough"

Symptoms:
- Low accuracy on your security tasks

- Too slow for interactive use
- Constantly running out of VRAM

Mitigation:
- Start with 8B model (faster iteration)
- Test quantization levels (Q4 vs Q5 vs Q8)
- Use model routing (8B for simple, 70B for complex)

Backup Plan: Hybrid approach (local for triage, cloud API for complex analysis)

Risk 3: "Management doesn't care about my chatbot"

Symptoms:
- No one uses your tool
- "We're too busy for this"
- No recognition or reward

Mitigation:
- Solve THEIR pain point, not yours (what do they complain about most?)
- Show metrics (30% ticket reduction = $X saved)
- Get executive sponsor (find someone who believes in AI)

Backup Plan: Build for external customers instead (go straight to commercialization)

Risk 4: "I get stuck on technical problems"

Symptoms:
- Can't debug OOM errors
- RAG returns irrelevant results
- Don't understand quantization tradeoffs

Mitigation:
- Use Ollama (simpler than vLLM/llama.cpp)
- Start small (100 docs, not 10,000)
- Measure before optimizing (don't guess)

Backup Plan: Use Claude/ChatGPT as debugging assistant (describe error, get specific fix)

Resources You'll Need

Hardware (Already Have)

✅ 198GB RAM
✅ 32 CPU cores
✅ NVIDIA GPU (specify model to determine VRAM)

Recommendation: Verify GPU model and VRAM with `nvidia-smi`
- If <24GB VRAM → Use 8B models primarily
- If 24-48GB VRAM → Can run 70B Q4 quantized
- If 80GB+ VRAM → Can run 70B Q8 or even FP16

Software (Free/Open Source)

✅ Ollama (easiest inference engine)
✅ ChromaDB (local vector database)
✅ sentence-transformers (local embeddings)
✅ Python 3.10+ with venv
✅ Your existing security docs (markdown, PDF, text)

Cost: $0 (all open source)

Optional Paid Tools (For Production)

- LangSmith (observability): $0-50/month
- Weights & Biases (eval tracking): $0-50/month
- Notion (project management): $0 (free tier sufficient)

Cost: $0-100/month optional

How to Use This Roadmap

Daily Workflow

1. **Morning (30 min)**: Read next section of current subject
2. **Evening (1-2 hours)**: Implement what you learned
3. **Weekend (4-6 hours)**: Deep work on current week's deliverable

Weekly Workflow

1. **Monday**: Start new subject, read 80/20 core section
2. **Tuesday-Thursday**: Implement core concepts (20 min reading, 1.5 hours coding each day)
3. **Friday**: Review week's progress, debug issues
4. **Saturday-Sunday**: Build deliverable, test thoroughly

When You're Stuck

1. **Re-read** the relevant subject's "Blind Spots" section (common mistakes)
2. **Test** with simpler example (reduce scope until it works)
3. **Measure** what's actually happening (don't guess)
4. **Ask** Claude for debugging help (paste error + context)

When You're Ahead of Schedule

1. **Add features** to current deliverable (don't rush to next week)
2. **Improve quality** (accuracy, speed, UX)
3. **Document** what you learned (for future you)
4. **Share** with colleagues (get feedback, build support)

Your Competitive Advantages

Advantage 1: Infrastructure Expertise

What Others Don't Have: 20 years of Linux, virtualization, networking
How It Helps: You can deploy, scale, and debug production systems
Your Edge: Most AI engineers can't set up a VM; you can architect entire infrastructures

Advantage 2: Security Domain Knowledge

What Others Don't Have: Deep understanding of attack patterns, compliance, incident response
How It Helps: You know what security teams actually need (not what AI people think they need)
Your Edge: You can build tools that solve real problems, not tech demos

Advantage 3: Portfolio-Driven Learning

What Others Don't Have: Immediate access to real security data and problems
How It Helps: Everything you build is portfolio-worthy (not toy projects)
Your Edge: Your GitHub becomes proof of production-grade skills

Advantage 4: Dual Market Position

What Others Don't Have: Can sell to employers AND customers
How It Helps: Multiple paths to monetization (job, consulting, products)
Your Edge: If one path fails, you have alternatives

Next Steps (What to Do RIGHT NOW)

Step 1: Verify Your Hardware (15 minutes)

```
# Check GPU model and VRAM
nvidia-smi

# Check RAM
free -h

# Check CPU
lscpu | grep "CPU(s)"
```

Expected Output:
- GPU: RTX 3090 (24GB), A100 (80GB), or similar
- RAM: 198GB available
- CPU: 32 cores confirmed

Step 2: Read Subject 9 (2-3 hours)

Focus Areas:
- Section 1: Model selection (which quantization for your GPU?)
- Section 3: Week 1-2 step-by-step guide
- Section 4: Common failure modes (what to watch for)

Step 3: Set Up Development Environment (1 hour)

```
# Create project directory
mkdir -p ~/ai-security-tools
cd ~/ai-security-tools

# Python venv
python3 -m venv venv
source venv/bin/activate

# Install dependencies
pip install ollama chromadb sentence-transformers pydantic
```

Step 4: Start Week 1, Process 1 (Next Document)

Deliverable: "Week_1_Action_Plan.md" (coming next)
Structure: Process → Session → Step (as you requested)
Time: Follow at your own pace (no strict deadlines)

Appendix A: Subject Cross-Reference

When working on a specific task, refer to these subject combinations:

Task: "Classify security incidents"

Primary: Subject 1 (Prompt Engineering)
Secondary: Subject 4 (Structured Outputs)
Supporting: Subject 7 (Evaluation)

Task: "Extract IOCs from logs"

Primary: Subject 4 (Structured Outputs)
Secondary: Subject 1 (Prompt Engineering)
Supporting: Subject 2 (Data Engineering for batch processing)

Task: "Build security knowledge base"

Primary: Subject 5 (RAG)
Secondary: Subject 2 (Data Engineering)
Supporting: Subject 9 (Local Embeddings)

Task: "Optimize inference speed"

Primary: Subject 3 (Cost/Resource Optimization)
Secondary: Subject 9 (Model Selection)
Supporting: Subject 6 (Observability for measuring)

Task: "Automate threat hunting"

Primary: Subject 8 (Agent Frameworks)
Secondary: Subject 5 (RAG for threat intel)
Supporting: Subject 1, 4 (Tool prompts and outputs)

Appendix B: Your Custom Security Use Cases

These are specific applications mapped from your background:

Use Case 1: Kerberos Attack Detection

Your Expertise: 20+ years with Kerberos, golden ticket attacks
AI Application: Automated ticket analysis for privilege escalation
Subjects: 1 (prompts), 4 (structured detection), 7 (evaluation)

Use Case 2: Incident Triage Chatbot

Your Pain Point: Too many low-value tickets
AI Application: User-facing assistant for common issues
Subjects: 9 (local deployment), 1 (prompts), 4 (structured responses)

Use Case 3: Security Playbook Assistant

Your Asset: Years of accumulated documentation
AI Application: RAG over internal playbooks and procedures
Subjects: 5 (RAG), 2 (data pipeline), 9 (local embeddings)

Use Case 4: Log Analysis Automation

Your Expertise: Parsing complex security logs
AI Application: Extract IOCs, classify threats, route alerts
Subjects: 4 (structured extraction), 2 (batch processing), 3 (optimization)

This roadmap is now your strategic compass. The next document (Week_1_Action_Plan.md) will be your tactical playbook.

Start with Subject 9 + Week 1 Action Plan → You'll have a working chatbot in 2-3 weeks.